

About IIT JAM

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Actually, What is IIT-JAM?

IIT- Indian Institute of Technology
JAM-Joint Admission Test for Masters

IIT-JAM-2027

Organizing Institute: IIT Karagpur

Test Papers

- **It is a Computer Based Test (CBT) in SEVEN Papers at the Undergraduate (UG) level.**

- **Mathematics**
- **Mathematical Statistics**
- **Chemistry**
- **Physics**
- **Geology**
- **Biotechnology**
- **Economics**

Career Opportunities

The JAM exam is conducted to provide admission to:

- **M. Sc.**
- **M. Sc. (Tech.)**
- **MS (Research)**
- **M. Sc.-M. Tech. Dual Degree**
- **Joint M. Sc.-Ph. D. Dual degree**

at the IITs for taking up science as a career option for the students.

- **Job Opportunities at IT Sector with handsome salary**

23 IITs

- **Candidates in merit list are eligible for admission to around 3000 seats in various Postgraduates (PG) at IITs.**

- **IIT Dharwad • IIT Delhi • IIT Madras • IIT Bombay • IIT Kanpur • IIT Roorkee • IIT Guwahatti • IIT Hyderabad • IIT Indore • IIT (BHU) Varanasi • IIT Dhanbad • IIT Bhubaneswar • IIT Gandhinagar • IIT Ropar • IIT Patna • IIT Mandi • IIT Jodhpur • IIT Tirupati • IIT Bhilai • IIT Goa • IIT Jammu • IIT Kharagpur • IISC Bangalore • Variuos NITs.....**

About IIT JAM-2027

- **Test Papers consists of (i) Multiple Choice Question (MCQ) (ii) Multiple selective Questions (MSQ) (iii) Numerical Answer type (NAT) Questions.**
- **Candidates may appear in ONE or TWO Test Papers. For two test Papers, the second paper must be carefully selected to avoid session clash.**

WHO CAN APPLY?

- **Candidates who have completed an Undergraduate(UG) degree or currently studying in the final year of Undergraduate(UG) programme are eligible to apply for JAM-2026 examination.**
- **Foreign Nationals with Indian degree are eligible to apply, subject to the policy of the Admitting Institute.**

Distribution of Marks

MCQ (1M) : 10Q $\frac{1}{3}$ rd Negative

MCQ (2M) : 20Q $\frac{2}{3}$ rd Negative

MSQ (2M) : 10Q No Negative

NAT(1M) : 10Q No Negative

NAT (2M) : 10Q No Negative

Total No. Questions=60

Total Marks=100

Time-3hr.

Application Fees

Gender/Category	One test paper	Two test papers
Female/SC/ST/PwD*	₹1000/-	₹1,350/-
All others	₹2000/-	₹2700/-

***PwD-Person with Benchmark Disability**

JAM Exam Centers

Particularly in KARNATAKA

- Belagavi
- Bengaluru North, Bengaluru South
- Hassan
- Hubballi/Dharwad
- Kalaburagi
- Mangaluru
- Mysuru
- Shivamogga

Last Three Year Cutoff

IIT JAM 2026 (Conducted by IIT Bombay)

- **GM-12.65**
- **OBC/EWS-11.38**
- **SC/ST/PWD=6.32**

IIT JAM 2025 (Conducted by IIT Delhi)

- **GM-19.9**
- **OBC/EWS-17.91**
- **SC/ST/PWD-9.95**

IIT JAM 2024 (Conducted by IIT Madras)

- **GM-11.22**
- **OBC/EWS-10.02**

For any enquiry

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IISC Bengaluru

Gmail: jam@iisc.ac.in

Ph. No: 8022932392

Previous Year Question papers:

Link: [**https://jam2026.iitb.ac.in/oldQP.html**](https://jam2026.iitb.ac.in/oldQP.html)

Present status of knowledge

- **What do you mean by group?**

Think about four axioms!!!

Field ($\mathbb{F}$)

Let $\mathbb{F}$ be a set containing atleast two elements and satisfying the following axioms:

With respect to addition (+)

- $(\mathbb{F}, +)$ is an abelian group

That is, $\forall a, b, c \in \mathbb{F}$ then

(i). $a + b \in \mathbb{F}$

(ii). $(a + b) + c = a + (b + c)$

(iii). There exist $0 \in \mathbb{F}$ such that $a + 0 = a$

(iv). $\forall a$, there exist $-a \in \mathbb{F}$ such that $a + (-a) = 0$

(v). $a + b = b + a$.

With respect to multiplication (.)

- $(\mathbb{F}, .)$ is commutative with unity
- Every non-zero element must have multiplicative inverse

then. $(\mathbb{F}, +, .)$ is called field.

Examples for field

- $(\mathbb{R}, +, \cdot)$
- $(\mathbb{C}, +, \cdot)$
- $(\mathbb{Q}, +, \cdot)$

Question: Give an example which is not a field?

Vectorspace $V(F)$

Let V be any non-empty set with its elements are called as vectors and $(\mathbb{F}, +, \cdot)$ be a field. We define two binary operations as follows

Vector addition

- **For all** $u, v \in V$, $u + v \in V$
- **Vector addition is also called as internal composition on V**

Scalar multiplication

- **For each** $u \in V$, $\alpha \in \mathbb{F}$ $\alpha \cdot u \in V$
- **Scalar multiplication is also called as external composition on V**

The algebraic structure $(V, +, \cdot)$ equipped with vector addition and scalar multiplication is called vector space over field $\mathbb{F}$ if the following conditions are satisfied.

- $(V, +)$ is abelian group

- For all, $\alpha, \beta \in \mathbb{F}$ and $u, v \in V$

(a) $(\alpha + \beta) \cdot u = \alpha \cdot u + \beta \cdot u$

(b) $\alpha \cdot (u + v) = \alpha \cdot u + \alpha \cdot v$

(c) $(\alpha \cdot \beta) u = \alpha (\beta \cdot u)$

(d) $1 \cdot u = u, \quad 1 \in \mathbb{F} \text{ (unity)}$

Then we call it as V is vectorspace over the field $\mathbb{F}$ and is denoted by $V(\mathbb{F})$

• **Question:** Check the vector addition and scalar multiplication for

(1). $\mathbb{R}(\mathbb{Q})$

(2). $\mathbb{R}(\mathbb{C})$

$\mathbb{R}(\mathbb{C})$ is not satisfies vector multiplication. Therefore $\mathbb{R}(\mathbb{C})$ is not forms a vectorspace.

Theorem 1: Every field is vectorspace over its subfield

Theorem 2: Every field is vectorspace over itself

Question: Which of the followings are forms vectorspace?

(a). $\mathbb{R}(\mathbb{R})$

(b). $\mathbb{R}(\mathbb{Q})$

(c). $\mathbb{R}(\mathbb{C})$

(d). $\mathbb{C}(\mathbb{R})$

(e). $\mathbb{Q}(\mathbb{R})$

(f). $\mathbb{C}(\mathbb{C})$

(g). $\mathbb{Q}(\mathbb{Q})$

Hint: Apply above Theorem 1 and 2.

How to check the given set is vectorspace or not?

Ans: Check whether they satisfy vector addition and scalar multiplication. If they fail to satisfy anyone of them, then they do not forms vectorspace.

Some well known vector spaces

1. Matrix space

$M_n(\mathbb{F}) = \{A = [a_{ij}] : a_{ij} \in \mathbb{R}, \text{ or } \mathbb{C}, \text{ or } \mathbb{Q}\}$ over the field $\mathbb{F} = \mathbb{R}, \text{ or } \mathbb{C}, \text{ or } \mathbb{Q}$

$M_n(\mathbb{F})$ denotes vectorspace of all $n \times n$ matrix

2. Polynomial space

$P_n(\mathbb{F}) = \{p(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n : a_i \in \mathbb{R}, \text{ or } \mathbb{C}, \text{ or } \mathbb{Q}\}$
over the field $\mathbb{F} = \mathbb{R}, \text{ or } \mathbb{C}, \text{ or } \mathbb{Q}$

$P_n(\mathbb{F})$ denotes vectorspace of all polynomials of degree atmost n .

3. Euclidean space

$$\mathbb{R}^n(\mathbb{R}) = \{(a_1, a_2, \dots, a_n) : a_i \in \mathbb{R}\}$$

$\mathbb{R}^n(\mathbb{R})$ denotes n -dimensional Euclidean space

Subspace

Definition: Let V be a vectorspace over the field $\mathbb{F}$, let W be any non-empty subset of V is said to be subspace of V over $\mathbb{F}$ if W itself a vectorspace over the same field $\mathbb{F}$.

Ex: $V = \{(a, b); a, b \in \mathbb{R}\}$ and $\mathbb{F} = \mathbb{R}$. Clearly $V(\mathbb{R})$ is a vectorspace.

Now $W = \{(a, b) : a = b, a \in \mathbb{R}\}$ over $\mathbb{R}$.

Then W is subspace of V .

Criteria to check subspace

A necessary and sufficient condition for a non-empty subset W of vectorspace V to be subspace of V over the same field $\mathbb{F}$ if

One step test

- **Vector addition and scalar multiplication.**

That is, For all $w_1, w_2 \in W$ and $\alpha \in \mathbb{F}$

- $w_1 + w_2 \in W$
- $\alpha w_1 \in W$

Two step test

For all $w_1, w_2 \in W$ and $\alpha_1, \alpha_2 \in \mathbb{F}$

- $\alpha_1 w_1 + \alpha_2 w_2 \in W$.

$\mathbb{R}^2(\mathbb{R}) = \{(a, b); a, b \in \mathbb{R}\}$ is a vector space.

Then which of the following forms subspace.

1. $W_1 = \{(a, b) : a + b = 0, a, b \in \mathbb{R}\}$
2. $W_2 = \{(a, b) : 2a - b = 0, a, b \in \mathbb{R}\}$
3. $W_3 = \{(a, b) : a + b = 1, a, b \in \mathbb{R}\}$
4. $W_4 = \{(a, b) : a^2 + b = 0, a, b \in \mathbb{R}\}$
5. $W_5 = \{(a, b) : a = 0, a \in \mathbb{R}\}$
6. $W_6 = \{(a, b) : a^2 - b^2 = 1, a, b \in \mathbb{R}\}$

Hint: Check vector addition and scalar multiplication

Ans:1, 2, 5.

Keypoints regarding subspace

Let V be a vectorspace and W be its subset then

- If additive identity i.e. $0 \notin W \implies W$ is not subspace of V .
- If W contains non-homogeneous linear restriction then it will not forms subspace.
- If W contains non-linear restrictions then it will not forms subspace.

$\mathbb{R}^3(\mathbb{R}) = \{(a, b, c); a, b, c \in \mathbb{R}\}$ is a vector space.

Then which of the following forms subspace.

1. $W_1 = \{(a, b, c) : a + b = 0, a, b \in \mathbb{R}\}$
2. $W_2 = \{(a, b, c) : a - 2b = 1, a, b \in \mathbb{R}\}$
3. $W_3 = \{(a, b, c) : a + b + c = 1, a, b, c \in \mathbb{R}\}$
4. $W_4 = \{(a, b, c) : a^2 + b^3 + c = 0, a, b, c \in \mathbb{R}\}$
5. $W_5 = \{(a, b, c) : a = b = c, a \in \mathbb{R}\}$
6. $W_6 = \{(a, b, c) : abc = 1, a, b, c \in \mathbb{R}\}$

Hint: Check vector addition and scalar multiplication

Ans:1, 5.

IIT JAM Previous Year Questions

Which of the followings are subspace of $\mathbb{R}^2(\mathbb{R})$?

1. $W_1 = \{(x, y) : x + y = 0\}$
2. $W_2 = \{(x, y) : x + y = 1\}$
3. $W_3 = \{(x, y) : x - y = 0\}$
4. $W_4 = \{(x, y) : x - y = 1\}$

Hint: Check vector addition and scalar multiplication

Ans:1, 3.

IIT JAM Previous Year Questions

Which of the followings are subspace of $\mathbb{R}^3(\mathbb{R})$?

1. $W_1 = \{(x, y, z) : x + y + z = 0, 2y + z = 1\}$
2. $W_2 = \{(x, y, z) : 2x + 3y + 3z = 0, y + x = 0\}$
3. $W_3 = \{(x, y, z) : x^3 + y^3 = 0\}$
4. $W_4 = \{(x, y, z) : x + y + z = 0, 2y + 3z = 0\}$

Hint: Check vector addition and scalar multiplication

Ans: 2, 4.

Which of the followings are subspace of $\mathbb{R}^3$?

1. $W_1 = \{(x, y, z) \in \mathbb{R}^3 : (y + z)^2 + (2x - 3y)^2 = 0\}$
2. $W_2 = \{(x, y, z) \in \mathbb{R}^3 : y \in \mathbb{Q}\}$
3. $W_3 = \{(x, y, z) \in \mathbb{R}^3 : yz = 0\}$
4. $W_4 = \{(x, y, z) \in \mathbb{R}^3 : x + 2y - 3z + 1 = 0\}$

Hint: Check vector addition and scalar multiplication and definition of subspace

Ans:1

IIT JAM Previous Year Questions

Which of the followings are subspace of $M_n(\mathbb{R})$?

1. $W_1 = \{A \in M_n(\mathbb{R}) : |A| = 0\}$
2. $W_2 = \{A \in M_n(\mathbb{R}) : \text{Tr}(A) = 0\}$
3. $W_3 = \{A \in M_n(\mathbb{R}) : \text{Rank}(A) = 1\}$
4. $W_4 = \{A \in M_n(\mathbb{R}) : A^T = A\}$

Hint: Check vector addition and scalar multiplication and try to construct counter examples

Ans:2, 4.

Rank of a matrix

Number of non zero rows in row reduced Echelon form.

Example

1. The rank of the matrix $\begin{bmatrix} 1 & 1 & 0 & -2 \\ 2 & 0 & 2 & 2 \\ 4 & 1 & 3 & 1 \end{bmatrix}$ is

- (A). 1
- (B). 2
- (C). 3
- (D). 4

Hint: Use row reduced Echelon form

Ans: (C)

IIT JAM-2022 (1 Mark)

1. The rank of the matrix $\begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$ with entries in

$\mathbb{R}$ is

- (A). 0
- (B). 2
- (C). 5
- (D). 4

Hint: Use row reduced Echelon form

Ans: (D)

Properties of rank of a matrix

If A is a matrix of order $m \times n$ then

- **$\text{Rank}(A) \leq \min(m, n)$**
- **If B be matrix of order $n \times p$ then $\text{Rank}(A) + \text{Rank}(B) - n \leq \text{Rank}(AB)$**
- **Suppose $m=n$ and $|A| \neq 0$ then $\text{Rank}(A)=n$.**
- **Suppose $m=n$ and $|A| = 0$ then $\text{Rank}(A) < n$.**

IIT JAM PYQ (2M)

If $P = [p_{ij}]$ be 50×50 matrix such that $p_{ij} = \min(i, j)$ where $i, j = 1, 2, \dots, 50$ then rank of P equals

- (A) 1
- (B) 2
- (C) 25
- (D) 50.

Hint: Use row operations

Ans: (A)

Eigenvalue and Eigenvector

- Let A be a square matrix of order n , there exist a non zero V and a scalar λ such that $AV = \lambda V$, where λ is called an eigenvalue and V is called an eigenvector of A .

IIT JAM-2023 (2 Mark)

4. Let $A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & -2 & 5 \\ 2 & 5 & -3 \end{bmatrix}$ then the eigenvalues of A are...

(A). -4, 3, -3

(B). 4, 3, 1

(C). 4, $-4 \pm \sqrt{13}$

(D). 4, $-2 \pm 2\sqrt{7}$

Hint: $\text{Trace}(A) = \text{sum of eigenvalues}$ and

$\text{Determinant}(A) = \text{product of eigenvalues of } A$

Ans: (C)

5. Let A be a 3×3 matrix with real entries such that $\det(A) = 6$ and the $\text{trace}(A) = 0$. If $\det(A + I) = 0$, where I denotes the 3×3 identity matrix, then the eigenvalues of A are....

- (A) -1, 2, 3
- (B) -1, 2, -3
- (C) 1, 2, -3
- (D) -1, -2, 3

Hint: $\text{Trace}(A)$ = sum of eigenvalues and
 $\text{Determinant}(A)$ = product of eigenvalues of A
Ans: (D)

IIT JAM-2023 (2 Mark)

7. Consider the matrix $A = \begin{bmatrix} 1 & 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & 4 & 4 & 4 \end{bmatrix}$

and B be a 5×5 real matrix such that AB is the zero matrix. Then the possible rank of B are equal to.....

- (A). 0
- (B). 1
- (C). 2
- (D). 3

Hint: Use properties of rank of a matrix

Result

If sum of each row (column) of a matrix $A_{n \times n}$ is k then k is one of the eigenvalue.

Infact that k is the largest eigenvalue of A .

Also, sum of the elements in the matrix $A_{n \times n}$ is nk .

IIT JAM-2021 (1 Mark)

Let $A = \begin{bmatrix} 2 & -1 & 3 \\ 2 & -1 & 3 \\ 3 & 2 & -1 \end{bmatrix}$. Then the largest eigenvalue of A is 4

Interesting Question

Q.2

To continue the sequence of tiles shown, the tile indicated by the question mark should be



(A)



(B)



(C)



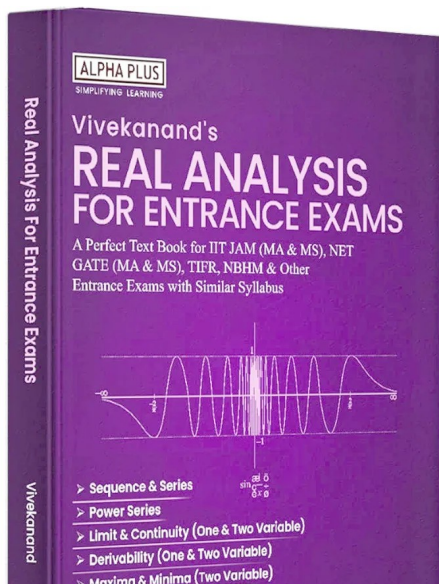
(D)

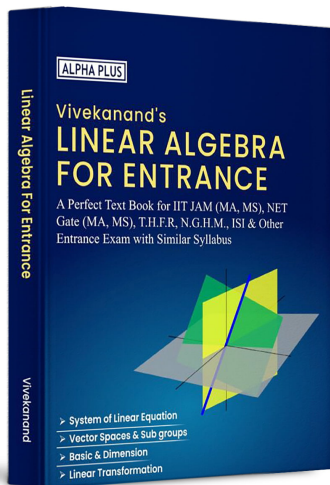


What Can You do now?

- **Collect Syllabus**
- **Collect Previous Year Question Paper and analyze it.**
- **Understand the concepts thoroughly.....meticulously....**
- **Solve MORE Problems/Questions**

Study Materials





Study Materials

